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A

Report on

SoC–Based Inverter Control Strategy for Grid-Connected Battery Energy Storage in AC Microgrid With Techno-Economical Analysis

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Abstract

This project presents the design, implementation, and techno-economic evaluation of a State of Charge (SOC)-aware inverter control strategy for grid-connected Battery Energy Storage Systems (BESS) operating within an AC microgrid. Conventional inverter droop control schemes regulate active and reactive power based strictly on grid-side voltage and frequency deviations, neglecting the battery's internal SOC and leading to cell over-discharge, accelerated degradation, and high replacement costs. To address this, a multi-layer control architecture is implemented in MATLAB/Simulink using a modified CIGRE low-voltage AC microgrid benchmark. The control scheme integrates a nested SOC-voltage droop loop into a Synchronous Reference Frame (SRF) *d-q* current controller. Simulation results validate that: Active power injection dynamically rolls off as the battery discharges below 85% SOC. Active and reactive power control channels remain decoupled under dynamic load step changes. A strict 35% SOC lower threshold automatically disconnects the inverter to prevent deep-discharge damage. A Power-Voltage (P-V) droop variant superiorly stabilizes local terminal voltages compared to standard P-Q droop control. A 20-year techno-economic evaluation for a representative 55 kW Solar PV + 210 kWh BESS microgrid demonstrates a Levelized Cost of Energy (LCOE) of NPR 17.58/kWh. By utilizing Time-of-Day (TOD) tariffs and substituting expensive diesel generator usage, the system achieves complete equity payback in Year 14 and cumulative net savings of NPR 7.04 Million over 20 years. The study confirms that preserving battery State of Health (SOH) via SOC-aware primary control is directly linked to guaranteeing long-term financial viability.

Table of contents

Acknowledgment i

Abstract ii

Table of contents iv

List of Figures v

List of Tables vi

Abbreviations vii

1 Introduction **1**

1.1 Background 1

1.2 Statement of Problem 2

1.3 Objective of Study 3

2 Literature Review **4**

3 Methodology **6**

3.1 Microgrid Modeling 6

3.2 BESS and Power Sharing Modeling 6

3.2.1 P–Q Control 6

3.2.2 P–V Control 7

3.3 Multi-Layer Power System Architecture 8

3.4 Simulation and Test Scenarios 8

3.5 Performance Evaluation & Benchmarking 9

3.6 Techno-Economic and LCOE Modeling in Solar-BESS Microgrids . 9

4 Results **10**

4.1 Simulink Validation and SOC Based Control System 10

4.2 Case-I to IV 11

4.2.1 Case I 11

4.2.2 Case-II 13

4.2.3 Case-III 14

4.2.4 Case-IV 14

4.3 LCOE 16

4.3.1 Financial 16

5 Conclusion and Future Works **19**

5.1 Conclusion 19

5.1.1 Dynamic Active Power Roll-Off 19
5.1.2 Decoupled Power Channel Performance 19
5.1.3 Over-Discharge Asset Protection 20
5.1.4 Power-Voltage (P-V) Droop Control 20
5.1.5 Levelized Cost and Energy Arbitrage Viability 20
5.1.6 Post-Debt Financial Recovery 21
5.1.7 Technical-Financial Control Coupling 21
5.2 Future Works 21
References **21**
Appendix . . . . . A

List of Figures

4.1 MATLAB Simulation of the CIGRE Model 10
4.2 BESS included in the CIGRE Model 10
4.3 SOC drop Vs Time 12
4.4 Drop in Active Power Vs Time 12
4.5 Reactive Power Injection 12
4.6 Increase in Grid Power 13
4.7 Step increase in Reactive Power 13
4.8 Effect of the reactive power on active power 14
4.9 Control of Inverter after reaching 35% SOC 14
4.10 Reactive Power Injection with BESS 1 and 2 15
4.11 PU Voltage graph during steady state 15

List of Tables

4.1 Components 11
4.2 LCOE for various conditions 16
4.3 Financial Projection 17

1 Specification . . . . . A
2 Cost Specification . . . . . A
3 Energy Consumption . . . . . B
4 Assumptions C

List of Abbreviations

AC: Alternating Current

BESS: Battery Energy Storage System

CapEx: Capital Expenditure

CIGRE: International Council on Large Electric Systems

DC: Direct Current

DG: Diesel Generator

DOD: Depth of Discharge

EMS: Energy Management System

EV: Electric Vehicles

GHG: Green House Gases

GTI: Grid–Tied Inverter

LCOE: Levelized Cost of Energy

MG: Microgrid

MW: Mega Watts

NEA: Nepal Electricity Authority

NPV: Net Present Value

OpEx: Operational Expenditure

P–Q: Active Power–Reactive Power (control)

P–V: Active Power–Voltage (control)

PCC: Point of Common Coupling

PI: Proportional-Integral

PLL: Phase-Locked Loop

PV: Photovoltaic

PWM: Pulse Width Modulation

RES: Renewable Energy Source

SOC: State of Charge

SOH: State of Health

SRF: Synchronous Reference Frame

WT: Wind Turbine

Chapter 1

Introduction

1.1 Background

Rapid advancements in battery manufacturing technologies and significant reductions in energy storage costs have catalyzed a global surge in the deployment of Battery Energy Storage Systems (BESS) across power networks. Concurrently, modern electrical grids and decentralized microgrids are experiencing an unprecedented penetration of non-dispatchable, intermittent renewable energy sources (RES), predominantly solar photovoltaic (PV) and wind generation. While RES integration is essential for decarbonization and energy security, the stochastic nature of solar irradiance and wind speed introduces severe power fluctuations, voltage unbalance, and frequency instability at the Point of Common Coupling (PCC). In this context, BESS serves as an indispensable dynamic buffer, absorbing excess renewable generation during low-demand periods and injecting active/reactive power during peak demand or unexpected generation shortfalls to ensure overall grid resilience.

In low-voltage Alternating Current (AC) microgrid architectures, grid-following inverters interface BESS units with the utility grid and local loads. Conventional inverter control frameworks predominantly employ fixed power-frequency (P-f) or power-voltage (P-V) droop schemes that mimic the inertial response of synchronous generators. However, traditional droop strategies regulate output power strictly based on grid-side voltage and frequency deviations without actively monitoring the internal state of the battery. Neglecting the State of Charge (SOC)—the primary metric defining usable stored energy—can lead to severe operational issues, including deep cell discharges, thermal stress, cell imbalances, and accelerated chemical degradation. To maximize battery calendar and cycle life, inverter controllers must adopt SOC-aware control topologies that dynamically scale active power injection according to real-time SOC levels and enforce strict lower bounds (e.g., a 35% SOC lower threshold) to prevent over-discharge.

Beyond technical stability, BESS deployment is heavily governed by techno-economic considerations. Despite ongoing price reductions, energy storage remains the most

capital-intensive asset within a hybrid microgrid. For instance, in a representative 55 kW Solar PV + 210 kWh BESS commercial/residential microgrid in Nepal, the battery storage system accounts for NPR 11.55 Million out of a total NPR 13.75 Million capital expenditure—representing approximately 84% of total project investment. Consequently, relying on BESS purely as an underutilized standby emergency backup yields poor financial returns. To achieve commercial viability, modern microgrid energy management systems (EMS) utilize BESS for multi-objective operational strategies, including Time-of-Day (TOD) tariff energy arbitrage—charging during low-cost off-peak hours (e.g., NPR 6.90/kWh during the wet season) and discharging during high-cost peak periods (e.g., NPR 12.60/kWh)—and fuel substitution for expensive diesel backup generators (NPR 30.00–60.00/kWh).

Ultimately, the technical performance of inverter control loops and the long-term financial feasibility of microgrids are intrinsically linked. Implementing an SOC-aware droop-regulated inverter control strategy preserves battery state-of-health (SOH) over its design lifetime (e.g., 6,000 cycles), avoiding premature capital replacement costs and enabling the microgrid to achieve a competitive Levelized Cost of Energy (NPR 17.58/kWh versus NPR 21.54/kWh without PV/BESS) alongside positive 20-year net cash flows. Therefore, developing robust, SOC-dependent droop control strategies for grid-tied BESS inverters is paramount for achieving both grid operational stability and long-term project bankability.

1.2 Statement of Problem

The models which are currently implemented in the BESS-based inverter only account for the load and its variation for the droop regulation. The grid-tied inverter (GTI) based inverter only follows the droop-regulated system. This hampers the system which contains the costly battery. Utilizing the BESS to its maximum capacity comes with its degradation of the lifespan. Just utilizing the BESS for maximum benefit cannot happen when it is solely used as a backup system.

Effective power sharing based on the SOC and its adaptation can help in the longevity of the battery and the overall system. However, this can reduce the

power output from the BESS device, but it can also be mitigated by the implementation of a smart EMS system. Also this can help to replace the fuel based emission source during the off-grid operation of the system.

1.3 Objective of Study

Main Objective

- To create a droop-regulated BESS-based system with an SOC-based inverter system for an AC microgrid.

Specific Objective

- To develop a control logic to be implemented in an inverter control
- To identify the parameters of the inverter based on the design specification
- To validate the droop control in the simulated environment
- To provide the techno-economical analysis by calculation of LCOE for the BESS over 20 years in operation in cooperating NEA tariff rate, discounted pay back period.

Chapter 2

Literature Review

Droop control is the most widely used method for autonomous, communication-free power sharing among inverter-interfaced sources in a microgrid, since it mimics the frequency- and voltage-dependent behaviour of synchronous generators [2]. Conventional droop schemes regulate active and reactive power output based on deviations of grid frequency and voltage from their nominal values, but a large fraction of the existing literature does not explicitly account for the state of charge (SOC) of the battery energy storage system (BESS) supplying the inverter, which can result in inefficient power management, accelerated battery degradation, and suboptimal system performance [3].

A number of studies have attempted to incorporate SOC into the droop framework, but predominantly for an islanded or DC microgrids. Wu et al. [4] proposed a droop strategy to achieve SOC balance among multiple energy storage units of varying capacity in an islanded AC microgrid. Su et al. [5] combined droop control with DC bus signaling to coordinate BESS operation in a DC microgrid without requiring a communication link, while Hoang and Lee [6] presented a voltage-restoration method that achieves accurate power sharing and SOC balancing among distributed battery units. Zeng et al. [7] and Belal et al. [8] proposed decentralised control strategies that adjust droop coefficients according to SOC and battery capacity to achieve SOC balance and bus-voltage stability in islanded DC microgrids, and Wang et al. [9] introduced a virtual-battery-based droop method that dynamically adjusts the droop coefficient and reference output voltage of the storage system based on its charge state.

Comparatively fewer works address SOC-aware control explicitly for AC microgrids, and most of these focus only on achieving SOC balance among multiple units rather than on optimising the energy utilisation of an individual BESS-fed inverter while it is exporting active and reactive power to the grid [10, 11]. Allabadi et al. [1] address this gap directly: they propose an SOC-based droop control strategy for a single grid-connected, BESS-fed inverter operating within a modified CIGRE low-voltage microgrid benchmark. Their scheme augments a conventional synchronous-reference-frame (SRF) current controller with an SOC-voltage droop

loop, so that the reference d-axis current supplied to the inner current controller is adjusted according to the deviation between the measured and reference SOC. The strategy is shown, through MATLAB/Simulink simulation, to reduce the inverter's active-power reference smoothly as the battery discharges, to disconnect the inverter safely when the SOC nears its lower protection limit, and to maintain stable operation under step changes in the reactive-power reference. The authors further compare this with a power–voltage (P–V) droop variant, which regulates the terminal voltage directly rather than reactive power, and report improved voltage regulation performance under varying SOC conditions.

This proposal builds directly on the work of Allabadi et al. [1], using their control formulation and benchmark system as the basis for an independent MATLAB/Simulink implementation, in order to validate the reported behavior and to serve as a foundation for further exploration of SOC-based control strategies for BESS-fed inverters in AC microgrids. Additionally, the project adds the financial aspect of the BESS injection into a microgrid. The work of Tamrakar et al. [12], provide with the insight on importance of the techno-economical analysis. The work provides information on the LCOE of the DG as Rs.30/kwhr. Along with this, NEA provides with the ToD tariff to its customer. The provided tariff is separated according to two season as dry and wet season. [13] Along with that there are various consequences which are related to the burning of the fossil fuel. The diesel generator which is utilizied during any outage of the grid can be replaced by the help of battery storage system. A single liter of diesel produces around 2.64 Kg of CO_2. [14]

Chapter 3

Methodology

The methodology follows the control strategy and benchmark system reported by Allabadi et al. [1], implemented independently in MATLAB/Simulink (Simscape Electrical, Specialized Power Systems library) to reproduce and validate the reported results.

3.1 Microgrid Modeling

A modified version of the CIGRE low-voltage microgrid (MG) benchmark is used as the test system. The network consists of a 0.4 kV, 50 Hz main feeder supplying six laterals with defined line resistance and inductance, four distributed loads (with a combined rating of 202 kVA), two photovoltaic (PV) sources, one wind turbine (WT), and two BESS units, each rated at 30 kW, connected through grid-following inverters. The line and load parameters are taken directly from the benchmark to allow like-for-like comparison of results.

3.2 BESS and Power Sharing Modeling

3.2.1 P–Q Control

Each BESS-fed inverter is controlled using a closed-loop current controller implemented in the synchronous reference frame (SRF). Measured three-phase voltages and currents are transformed to the *dq*-frame using Clarke and Park transformations referenced to a phase-locked loop (PLL). Measured active and reactive power (P_m, Q_m) are compared against dispatched references (P_{ref}, Q_{ref}) through an outer PI controller to generate reference *dq*-currents. A second, inner PI controller compares these reference currents against the measured *dq*-currents to generate the reference voltage, which is transformed back to the *abc*-frame and passed to the PWM generator. Standard active power-frequency (P-f) and reactive power-voltage (Q-V) droop relations are expressed as:

$$f = f_0 - m_p(P_m - P_{ref})$$

$$V = V_0 - n_q(Q_m - Q_{ref})$$

The SOC-based droop loop is incorporated as follows. The deviation between the reference SOC and the measured SOC of the BESS ($\Delta SOC = SOC_{ref} - SOC_m$) is scaled by an SOC-voltage droop coefficient k_{SoC-V} to obtain a voltage correction term ΔV, which is in turn scaled by a coefficient k_{V-I} to produce a correction current ΔI. This correction is added to the reference d-axis current before it is passed to the inner current-control PI loop, so that the active power delivered by the inverter is automatically reduced as the battery discharges, and increased toward the nominal set point as the SOC approaches its reference value. As modeled in the modified CIGRE testbed, the strategy augments a Synchronous Reference Frame (SRF) d-q current controller with a nested SOC-voltage droop loop:

$$\Delta\text{SOC} = \text{SOC}_{ref} - \text{SOC}_m$$

$$\Delta V = k_{\text{SoC-V}} \cdot \Delta\text{SOC}$$

$$\Delta I = k_{\text{V-I}} \cdot \Delta V$$

A lower SOC protection limit (35%) triggers disconnection of the inverter to prevent over-discharge, and an upper limit (85%) is used to hold the nominal power reference constant.

3.2.2 P–V Control

The instantaneous magnitude of the Point of Common Coupling (PCC) or inverter terminal voltage (V_m) is derived from orthogonal stationary components:

$$V_m = \sqrt{V_\alpha^2 + V_\beta^2}$$

The error between the predetermined voltage reference (V_{ref}) and the measured terminal voltage (V_m) is passed through an outer Proportional-Integral (PI) voltage regulator to directly determine the quadrature-axis reference current ($I_{q,ref}$):

$$I_{q,ref} = \left(K_p + \frac{K_i}{s}\right)(V_{ref} - V_m)$$

Where:K_p and K_i represent proportional and integral gains of the terminal voltage

controller. $I_{q,ref}$ dictates the reactive current injected or absorbed by the inverter to restore $V_m \approx V_{ref}$

3.3 Multi-Layer Power System Architecture

The overall control architecture is organized in three layers: (i) an upper dispatch layer, which sets P_{ref}, Q_{ref}, and SOC_{ref} for each BESS unit; (ii) a middle SOC-droop layer, which adjusts the d-axis current reference according to the measured SOC; and (iii) a lower inner-loop control layer, which regulates the inverter's injected current and voltage to track the adjusted references. This layered structure keeps the SOC-awareness logic modular, so that it can later be extended or replaced (e.g., with a power–voltage variant) without altering the inner current-control loop.

3.4 Simulation and Test Scenarios

The following case studies, mirroring those used to validate the original strategy, will be simulated:

- **Case I – Fixed power references:** Both BESS-fed inverters operate at fixed $P_{ref} = 30$ kW and $Q_{ref} = 4$ kVAR, starting from different initial SOC values, to observe the active-power roll-off as SOC declines.
- **Case II – Step change in reactive power reference:** A step change in Q_{ref} is applied to one inverter to evaluate the dynamic response and decoupling between active and reactive power control.
- **Case III – Near-depleted battery:** Initial SOC values are set close to the lower protection limit to validate the automatic disconnection behavior at $SOC = 35\%$.
- **P–V control comparison:** The SOC-based P–Q droop strategy is compared against a power–voltage (P–V) droop variant, in which the inverter regulates its output voltage directly rather than reactive power.

3.5 Performance Evaluation & Benchmarking

The simulated results (active and reactive power profiles, SOC trajectories, terminal voltage, and current waveforms) will be bench marked against the results reported by Allabadi et al. [1] for the same test cases, to validate the correctness of the independent implementation. Performance will be evaluated in terms of: power-reference tracking accuracy, voltage regulation under the P–V strategy versus the P–Q strategy, and the effectiveness of the SOC-droop mechanism in preventing over-discharge while maintaining uninterrupted supply to the loads (via grid support).

3.6 Techno-Economic and LCOE Modeling in Solar-BESS Microgrids

While technical control strategies ensure grid stability and lifespan preservation, BESS deployment must be justified economically due to its high capital expenditure. In hybrid solar PV + BESS microgrids, economic evaluation relies on Levelized Cost of Energy (LCOE), Net Present Value (NPV), and discounted payback metrics. The microgrid Levelized Cost of Energy ($LCOE_{MG}$) evaluates the lifetime unit cost of generated/serviced electricity:

$$\mathrm{LCOE_{MG}} = \frac{\mathrm{CapEx_0} + \sum_{t=1}^{N} \frac{\mathrm{OpEx}_t + \mathrm{Debt\ Service}_t}{(1+r)^t}}{\sum_{t=1}^{N} \frac{E_t}{(1+r)^t}}$$

where $\mathrm{CapEx_0}$ is the total initial capital outlay, OpEx_t represents annual operation and maintenance expenses, $\mathrm{Debt\ Service}_t$ is loan principal and interest payments, E_t is total energy delivered, r is the discount rate (e.g., 10%), and N is project lifespan (20 years).

Chapter 4

Results

4.1 Simulink Validation and SOC Based Control System

A validated MATLAB/Simulink model of the modified CIGRE low-voltage AC microgrid benchmark, incorporating two BESS-fed grid-connected inverters, PV, and wind-turbine sources. Figure 4.1 provides a basic overview of the system. Wind turbine and PV plant are included as a PV bus, and the grid is taken as a slack bus. Load are included as a series RLC load. Figure 4.2 gives the structure of the inverter implemented in this project. It has a simple LCL circuit with a SOC-based control.

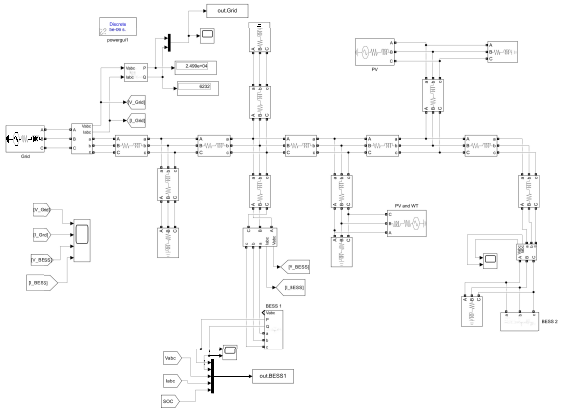


Figure 4.1: MATLAB Simulation of the CIGRE Model

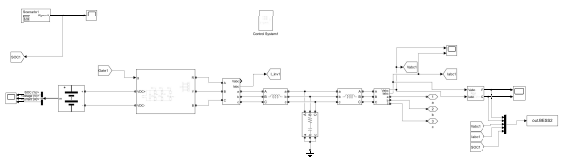


Figure 4.2: BESS included in the CIGRE Model

Components	Value	Unit
Inductor	5.1m	H
Capacitor	95u	F
Capacitor(DC)	270u	F
Voltage line to line	400	V
Voltage(DC)	800	V
Frequency	50	Hz
Sampling Frequency	10k	Hz

Table 4.1: Components

An independently implemented SOC-based droop control strategy for the BESS-fed inverter, including the outer power-control loop, inner current-control loop, and the SOC-voltage droop augmentation. The table **??** provides the values set for various components of the inverter.

4.2 Case-I to IV

Simulation results for Cases I–III (fixed power reference, step change in reactive power reference, and near-depleted battery disconnection), bench marked against the results reported in the reference study. Figure 4.3 shows that there is linear drop of the SOC level, this was created to analyze the system performance under various SOC in quick interval. The implemented battery was of 800 V and 30 Ah. To show the effectiveness of the system, implementation of the linear drop of SOC was performed.

4.2.1 Case I

Based on this Case-I can be observed in the figure 4.4. Since the SOC level in the BESS 1 was $> 85\%$. It was providing rated power, but the BESS 2 was already at 80% SOC it had a drop in the power injection. Similarly, when the SOC dropped below 85% in BESS 1, it then gradually dropped its power injection.

A constant reactive power is to be supplied in the system by the BESS, as

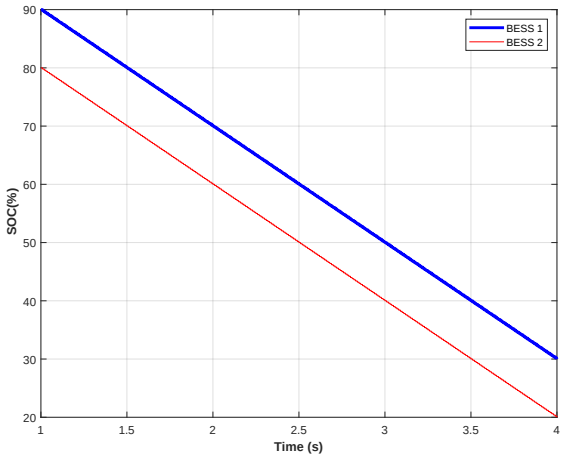


Figure 4.3: SOC drop Vs Time

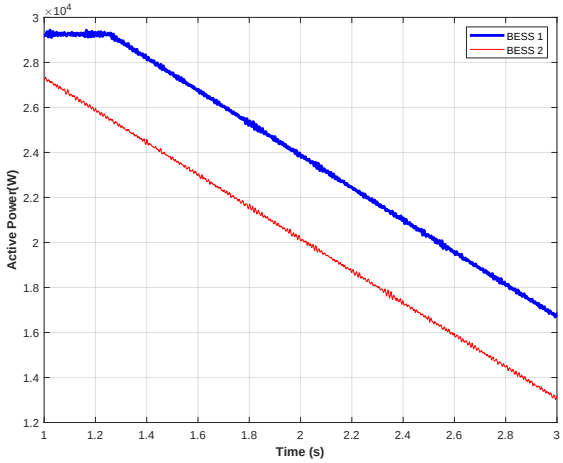


Figure 4.4: Drop in Active Power Vs Time

observed in the figure 4.5. Reactive power injected to the grid was set to around 4.5 kVAr. Finally, the observation 4.6 of the grid power injection shows that

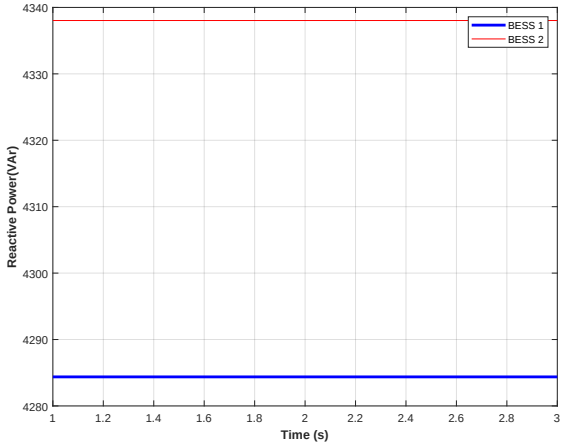


Figure 4.5: Reactive Power Injection

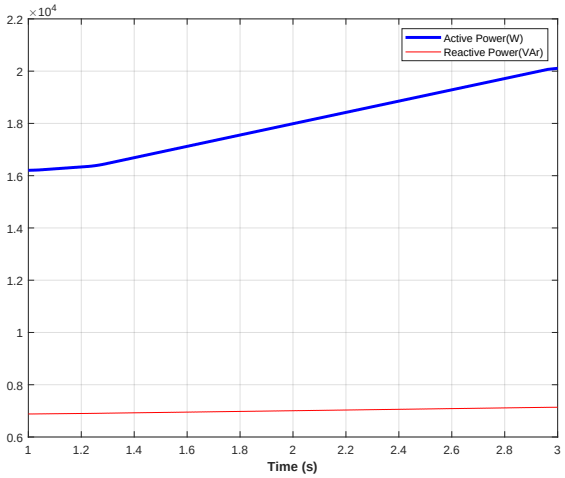


Figure 4.6: Increase in Grid Power

decrease in the BESS power injection increases the grid power.

4.2.2 Case-II

A step increase in reactive power is performed in the network. The constant reactive power is increased to 15kVAr from BESS 2. To perform the analysis regarding the effect of the injection of reactive power to the overall system. Figure 4.7 shows the injection of reactive power to the grid. Similarly, active power shows

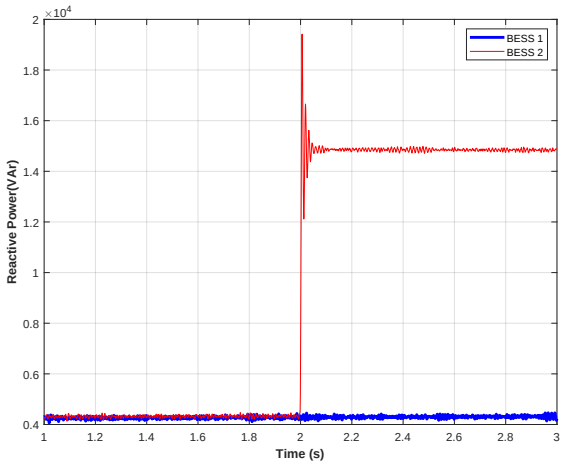


Figure 4.7: Step increase in Reactive Power

a transient during the injection of reactive power from the BESS to the grid as observed in the simulation result.

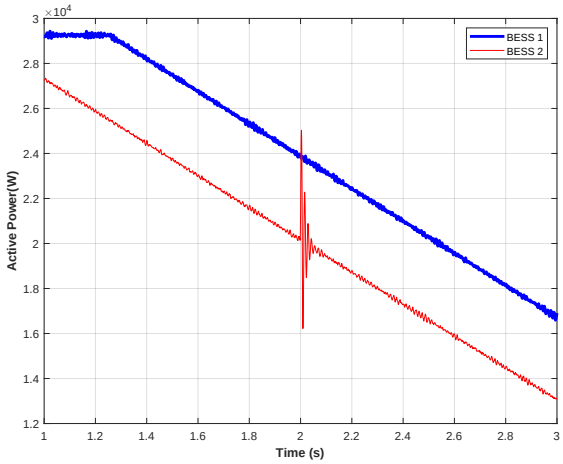


Figure 4.8: Effect of the reactive power on active power

4.2.3 Case-III

The lower SOC protection limit is configured at 35% to prevent battery degradation. When the SOC of BESS 2 reaches 35 % (at $t \approx 3.25$ s), the control loop automatically triggers inverter disconnection, dropping active power injection to 0 W. Similarly, BESS 1 disconnects when its SOC reaches 35% (at $t \approx 3.75$ s), successfully validating the over-discharge protection mechanism.

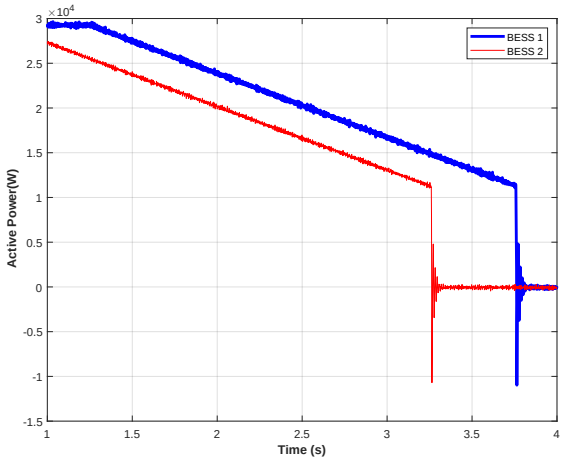


Figure 4.9: Control of Inverter after reaching 35% SOC

4.2.4 Case-IV

To address the limitations of conventional P-Q droop control in maintaining bus voltage stability under declining State of Charge (SOC) levels, a Power-Voltage (P-V) droop control scheme was implemented in the BESS-fed grid-connected

inverters. Unlike the P-Q strategy where reactive power injection remains fixed at a dispatched setpoint (Q_{ref}), the P-V scheme continuously regulates reactive power output to maintain the inverter terminal voltage (V_m) at a pre-determined reference setpoint (V_{ref}).

As shown in Figure 4.10, as the SOC of BESS 1 and BESS 2 declined linearly over time (with active power injection rolling off accordingly from 30 kW and 25 kW), the P-V controller automatically ramped up reactive power injection. BESS 1 reactive power increased dynamically from ~ 0.9 kVAR to over 4.2 kVAR at $t = 3.0$ s, while BESS 2 increased its reactive power injection from ~ 0.75 kVAR to 1.35 kVAR. This benchmark confirms that the P-V droop strategy significantly enhances local voltage regulation and eliminates the need for external reactive power compensation devices in AC microgrids with dynamic SOC variation.

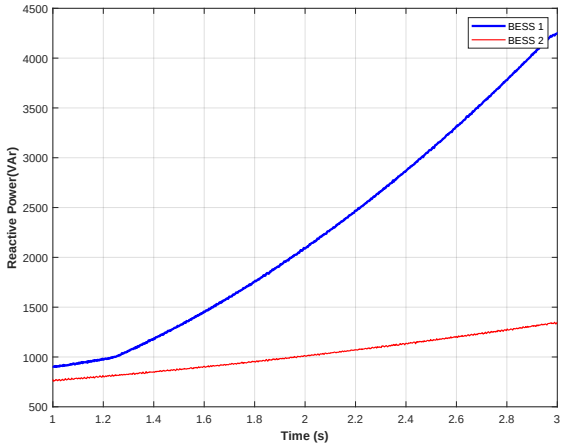


Figure 4.10: Reactive Power Injection with BESS 1 and 2

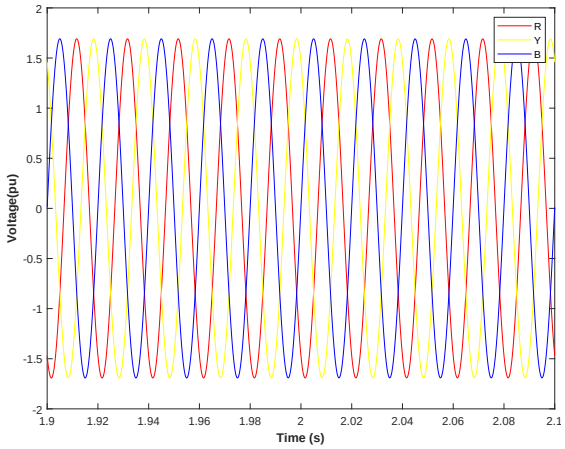


Figure 4.11: PU Voltage graph during steady state

4.3 LCOE

The paper provides with the information that it would help the system to maintain the LCOE in the overall system. It also claims the benefit of BESS injection by ensuring efficient energy use. In this section, with the help of real energy consumption, LCOE is determined, with estimated value of solar generation. Renewable energy is essential for minimizing the GHGs and carbon emissions. However due to intermittent, unpredictable nature of solar energy, it is essential for the addition of the BESS. But, the cost aspect of the system integration is also essential for the actual viability. In this section, cost analysis is performed for the BESS and solar integration in a residential building. The detail of the energy consumption is mentioned below in the table 3

LCOE for Grid with Diesel Generator	NPR 21.54
LCOE for Grid	NPR 12.19
LCOE(Capex + Cost)/Energy Output	NPR 17.58
Total LCOE with Grid+BESS+PV	NPR 29.77
Diesel energy offset for 20 years	300,000.00
Diesel Fuel Saved(litre)	50,000
Total equivalent Carbon curbed(mt)	132
Total Energy remaining	71053.8
Cycle remaining after 20th year	338.3513137

Table 4.2: LCOE for various conditions

4.3.1 Financial

The table 1 below provides the information on the overall cost for the implementation of the solar and BESS system. The assumptions are based on the following criteria:

- The roof space provided for the solar system installation, the cost for the application of the solar system is based on the current market rate.

- The energy consumption during peak hours to be reduced to half.
- The approach to eliminate the use of diesel generator for the reduction of GHG emissions.
- The calculation is done with the total of 85% SOC to 35% SOC during the AC microgrid operation, to double the lifespan, and implement the system during outage to remove diesel generator operation.
- The cost for the kWhr of the battery is based on the present rate of battery including the shipping, tax and duty imposed in the BESS. The current market price of the battery is \$0.25/Wh.

Project Year	**Inflows (NPR)**	**Debt Service & Outflows (NPR)**	**Net Cash Flow (NPR)**	**Cumulative Net Savings (NPR)**
Year 0	0.00	2,750,000.00 (Equity)	−2,750,000.00	−2,750,000.00
Year 1	2,800,985.75	3,007,767.09	−206,781.34	−2,956,781.34
Year 2	1,343,135.46	3,007,767.09	−1,664,631.63	−4,621,412.97
Year 3	1,307,808.67	3,007,767.10	−1,699,958.43	−6,321,371.40
Year 4	1,278,542.80	3,007,767.10	−1,729,224.30	−8,050,595.70
Year 5	1,254,470.13	3,007,767.09	−1,753,296.96	−9,803,892.66
Year 6	1,234,853.53	71,500.00 (O&M)	+1,163,353.53	−8,640,539.13
Year 10	1,189,759.51	71,500.00	+1,118,259.51	−4,114,196.82
Year 14	1,179,878.69	71,500.00	+1,108,378.69	**+325,941.89**
Year 20	1,201,420.02	71,500.00	+1,129,920.02	**+7,035,653.66**

Table 4.3: Financial Projection

During Years 1–5, annual debt servicing (NPR 2,936,267.09/year) exceeds operational inflows, resulting in net annual cash outflows. However, following loan retirement in Year 5, the microgrid generates a steady net operational cash inflow

of $\sim$ NPR 1.12 Million / year. The initial equity investment (NPR 2.75 Million) reaches full financial payback in Year 14 (13.7 years), yielding a cumulative net profit of NPR 7,035,653.66 by Year 20.

A critical finding from the integrated techno-economic evaluation is the direct link between inverter primary control and project financial return. The 210 kWh BESS array represents 84% (NPR 11.55 Million) of the total system investment. The BESS manufacturer rates the pack for $6,000$ equivalent full cycles ($1,008,000$ kWh lifetime throughput). Over the 20-year project lifetime, total BESS energy throughput reaches $636,946.2$ kWh (equivalent to $3,791$ cycles). By enforcing the 35% lower SOC protection threshold within the control loop, the inverter prevents over-discharge and cell voltage imbalances. This preserves battery state-of-health (SOH) across its 20-year deployment, avoiding premature BESS replacement costs (NPR 8M–11.5M) and ensuring the NPR 7.04 Million cumulative net savings are successfully realized.

Chapter 5

Conclusion and Future Works

5.1 Conclusion

This study presented an independent validation and techno-economic evaluation of a State of Charge (SOC)-aware current-decoupled control strategy for grid-connected Battery Energy Storage Systems (BESS) operating within a modified CIGRE low-voltage AC microgrid benchmark. By integrating a nested SOC-voltage droop loop into a Synchronous Reference Frame (SRF) *d-q* current controller, the control topology successfully addresses the limitations of conventional fixed-gain droop schemes that fail to monitor battery energy reserves.

The primary conclusions derived from the MATLAB/Simulink simulations and techno-economic modeling are as follows:

5.1.1 Dynamic Active Power Roll-Off

The controller effectively regulates active power injection according to stored energy reserves. While BESS SOC remains above the 85% upper limit, the inverter delivers full rated power (30 kW). Once SOC falls below 85%, the active power output scales down linearly in direct proportion to SOC depletion, facilitating smooth power transition to the main utility grid without inducing frequency or voltage instabilities.

5.1.2 Decoupled Power Channel Performance

Under dynamic step changes in reactive power demand (stepping from 4.3 kVAR to 15 kVAR), the inner *d-q* current controller demonstrated rapid response tracking (settling time < 50 ms) while maintaining the linear active power roll-off trajectory. This confirms effective decoupling between *d*-axis (active power / SOC control) and *q*-axis (reactive power control) regulation.

5.1.3 Over-Discharge Asset Protection

Enforcing a hard lower protection threshold at 35% SOC successfully triggered automatic inverter disconnection, halting active power injection immediately. This mechanism reliably safeguards battery state-of-health (SOH) against irreversible deep-discharge degradation and cell imbalances.

5.1.4 Power-Voltage (P-V) Droop Control

The implementation of the Power-Voltage (P-V) control scheme in Case-IV successfully validated the advantages of dynamic reactive power support over standard P-Q droop regulation. By assigning fixed terminal voltage setpoints (V_{ref} = 0.9910 p.u. and 0.9812 p.u.) and allowing the outer PI loop to dynamically adjust q-axis reference currents ($I_{q,ref}$), the inverters ramped up reactive power output (reaching 4.2 kVAR for BESS 1 and 1.35 kVAR for BESS 2) as active power rolled off with declining SOC. This dynamic reactive power injection maintained stable bus voltages across the entire discharge cycle, whereas conventional P-Q control exhibited noticeable voltage degradation. The P-V scheme provides superior voltage stability and power quality for microgrids operating under varying load profiles and intermittent renewable energy penetration.

5.1.5 Levelized Cost and Energy Arbitrage Viability

The 20-year techno-economic evaluation for a representative 55 kW Solar PV + 210 kWh BESS residential microgrid yielded a Levelized Cost of Energy (LCOE) of NPR 17.58 / kWh, representing an 18.4% reduction compared to a baseline grid-less supply (NPR 21.54 / kWh). By leveraging Nepal Electricity Authority (NEA) Time-of-Day (TOD) tariffs—charging BESS during off-peak hours (NPR 6.90 / kWh) and discharging during peak evening demand (NPR 12.60 / kWh)—the system captures NPR 534,999.75 in annual energy value while displacing $300,000$ kWh ($50,000$ liters) of high-cost diesel generation over 20 years.

5.1.6 Post-Debt Financial Recovery

Despite initial net operational cash outflows during the 5-year debt servicing period (NPR 11.0M loan at 12% interest), loan retirement in Year 5 establishes a steady post-debt net operational inflow of $\sim$ NPR 1.12 Million / year. Full equity payback (NPR 2.75M initial outlay) is achieved in Year 14 (13.7 years), generating a cumulative net savings of NPR 7,035,653.66 by Year 20.

5.1.7 Technical-Financial Control Coupling

Because battery storage accounts for 84% (NPR 11.55M) of total project CapEx, enforcing the 35% SOC disconnection threshold directly preserves battery SOH across its $6,000$-cycle rated lifespan ($636,946.2$ kWh 20-year throughput). This control safeguard eliminates premature battery replacement outlays (NPR 8M–11.5M), guaranteeing that project financial yields and net savings are fully realized.

5.2 Future Works

- **Multi-BESS Adaptive SOC Balancing:** Extend the single-unit SOC droop loop to a multi-BESS secondary consensus layer that dynamically adjusts individual inverter droop coefficients ($k_{\text{SoC-V}}$) based on heterogeneous battery capacities and real-time SOH discrepancies.

- **Hardware-in-the-Loop (HIL) Experimental Validation:** Validate the Simulink controller models on a real-time digital simulator interfaced with a physical inverter testbed to analyze switching losses, sensor noise sensitivity, and micro-grid phase-locked loop (PLL) synchronization dynamics.

- **Harmonics Analysis:** During the operation of the inverter under its rated condition, due to the design considerations, filters are usually can be able inject unwanted harmonics into the system. This can cause various issues related to current harmonics, these are yet to be understood and can be the topic for the further analysis.

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Appendix

System and Finance Specification		**Consumption Specification**	
Cost per kW	NPR 40,000.00	Total Production/kW	1100
Solar PV size(kw)	55	Total Production(kWh)	55000
Cost of the solar system	NPR 2,200,000.00	Grid(Import)(kWh)	113000
		Grid(Export)(kWh)	14600
Cost per kWhr	NPR 55,000.00	Diesel(kWh)	15000
BESS Size(kWhr)	210	BESS(Offset)	38325
Cost of BESS	NPR 11,550,000.00	BESS Rated Cycle	6000
Total cost of project	NPR 13,750,000.00	Total energy in the BESS(kWh) through its life	1008000

Table 1: Specification

Cost Specification	
O&M	0.02%
Insurance	0.50%
Depriciation	
Year 1	50%
Year 2 onwards	15%

Table 2: Cost Specification

Apartment Montly Billing by NEA and Total Consumption					
Month/Year	**Peak(kWh)**	**Normal(kWh)**	**Offpeak(kWh)**	**Total Consumption**	**Total Energy utilization**
Sharwan/2080	2767.61	1777.78	2516.29	7061.68	7061.68
Bhadra/2080	2865.46	1969.36	2462.73	7297.55	14125.55
Ashwin/2080	3394.88	2112.53	2456.55	7963.96	10480.26
Karthik/2080	2072.36	1285.44	1464.66	4822.46	7041.16
Mangshir/2080	2498.78	2767.61	1965.24	7231.63	8745.03
Poush/2080	4608.22	4905.89	4243.6	13757.71	16733.98
Magh/2080	5622.77	5515.81	4924.43	16063.01	19119.41
Falgun/2080	4120	6362.31	5449.73	15932.04	20856.142
Chaitra/2080	1934.34	3217.72	3167.25	8319.31	14571.611
Baishak/2081	1571	1188	3199	5958	12912
Jestha/2081	1719.07	1577.96	4318.79	7615.82	12494.5
Ashar /2081	2139.31	2041.46	4924.43	9105.2	15234.41

Table 3: Energy Consumption

Energy Cost Specifications(Baishakh to Mangsir)	
Grid Cost (NPR/kWh)	12.6
Grid Cost Normal(NPR/kWh)	11.1
Grid Cost Offpeak (NPR/kWh)	6.9
Grid Export (NPR/kWh)	5.94
Demand Charge(NPR/kVA)	315
Energy Cost Specifications(Kartik to Chaitra)	
Grid Cost (NPR/kWh)	12.6
Grid Cost Normal(NPR/kWh)	11.1
Grid Cost Offpeak (NPR/kWh)	11.1
Grid Export (NPR/kWh)	5.94
Diesel Cost (NPR/kWh)	30
Diesel Escalation (%/yr)	0.02
Solar Degradation (%/yr)	0.006
BESS Degradation (%/yr)	0.02
Consumption Rise(%/yr)	0.015
Finance Structure	
Equity	20%
Debt	80%
Equity(NPR)	2,750,000.00
Debt(NPR)	11,000,000.00
Debt term(years)	5
No. of installment per year	12
Interest	12%
Discount rate	8%

Table 4: Assumptions